



Read and Change a Loop

Ontario Math C3.2 — read & alter code

Name: _____

Date: _____

★ I can read a loop, change it, and say what happens.

Read the code first

Good coders READ code before they change it — like reading a recipe before you cook.

Here is one of Bit's loops. Read it, see what it does, then change a number and predict the new result.

Read this loop

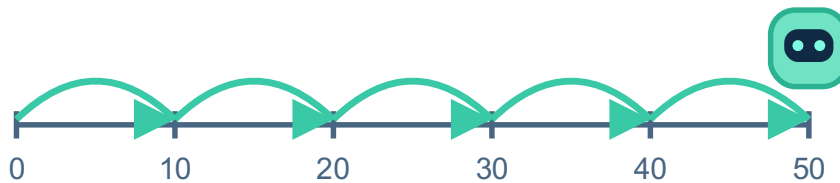
when green flag clicked

repeat 5

move 10 steps

Bit moves 10 steps, 5 times. That is 5 moves and 50 steps in total.

repeat $5 \times \text{move } 10 = 50 \text{ steps}$



Read it as 5 jumps of 10 → 50 steps.

1 Read it

Look at the loop above. How many times does Bit move? _____ Each move is 10 steps, so how far in TOTAL? _____

2 Change the count, predict

Change repeat 5 to repeat 3. PREDICT: how many moves now, and how far in total? Then check.

3 Change the inside

Now keep repeat 5, but change "move 10" to "move 2". How far does Bit go in total now? How did changing the inside block change the result?

4 Challenge

You want Bit to move 40 steps in total, moving 10 each time. What repeat number do you need? Show how you worked it out.

Big idea

Read a loop to see what it does. Changing the repeat number changes how many times; changing the inside block changes what happens each time.